

Masrur Sabab Khan Chowdhury

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Career Objective

As a curious and driven learner, I seek to explore diverse sectors of technology and innovation. I aim to apply my skills in programming, AI and data-driven analysis to solve real-world challenges. I am passionate about creating meaningful impact through technology while continuously growing as a problem solver. I believe in lifelong learning, teamwork and using innovation to improve lives and industries.

Education

- Southeast University** - Bachelor of Science in Computer Science and Engineering 2022 - 2026
Dhaka | CGPA - 3.84 / 4.00
- Ispahani Public School & College** - Higher Secondary Certificate (HSC) 2018 - 2020
Science | Cumilla Board | GPA - 5.00 / 5.00
- Islami Adarsha High School** - Secondary School Certificate (SSC) 2018
Science | Cumilla Board | GPA - 5.00 / 5.00

Professional Experience

Summit Communications Limited (Network Planning & Design Department) - Intern Nov 2025 - Jan 2026

- Developed a Datacom network visualization & modeling dashboard using Django (Python) to transform raw network data into interactive topology insights supporting network planning and design decisions.
- Implemented scalable topology graph visualization with Cytoscape.js, enabling interactive exploration (search, filtering, host-link analysis, layout algorithms) across complex, high-density network graphs.
- Built a GIS-based network mapping module using Leaflet.js, visualizing POP/sites with latitude-longitude coordinates, supporting map layer switching, and integrating KML/KMZ overlays for site-level geographic interpretation.
- Integrated and managed network datasets in MariaDB/MySQL (hosts, links, interfaces, capacity/bandwidth, utilization), optimized performance using DB connection pooling, TTL caching and collaborated with network engineers to validate topology accuracy and support deployment/integration.

Academic Projects

- BekarRank - Job Posting & Application App (Java, JavaFX, MySQL)** Sep 2023 - Jan 2024
 - Designed and developed a job posting and application platform with secure authentication.
 - Built an intuitive JavaFX interface with MySQL integration for seamless user experience.
 - Implemented CRUD operations, search and filter functionalities for efficient data handling.
- Thread & Style - E-commerce Website (HTML, CSS, PHP, MySQL, JavaScript)** Feb 2025 - Jun 2025
 - Built a dynamic online store with product filters, session-based cart and admin control panel.
 - Managed database-driven inventory using MySQL and ensured a modular, responsive UI.
 - Focused on user experience and smooth client-server interaction through PHP and JavaScript.
- US Road-Accident Analytics - ETL & Data Warehouse (Pentaho, SQL, Python)** Jul 2025 - Oct 2025
 - Processed and cleaned US accident data using Pentaho to build an analytical data warehouse.
 - Designed a star schema to support multi-dimensional analysis and visualization of accident trends.

- Applied ML models such as Decision Tree and Random Forest using Python for accident prediction and pattern detection.
- **Nagar Paribahan (Figma prototype)** Sep 2024 - Jan 2025
 - Designed a smart trip planning interface with route mapping, stops and payment flow.
 - Focused on user-friendly navigation and integration with Rapid Pass for ticketing.
 - Created high-fidelity wireframes to visualize user journeys and system flow.
- **Space-Shooter Game (C++, OpenGL)** Jul 2024 - Oct 2024
 - Developed a 2D shooting game with sprite animations, collision detection and scoring system.
 - Applied object-oriented design for game logic and real-time rendering in OpenGL.
- **10-bit ALU Design (Logisim)** Feb 2024 - Jun 2024
 - Designed and simulated a 10-bit Arithmetic Logic Unit with a custom Instruction Set Architecture (ISA) to supporting multiple operations.
 - Implemented add/subtract, AND/OR/XOR, shift and compare functionalities with test benches.

Academic Research

- **A Vision-Language Transformer Framework for Automated Radiology Report Generation (CNN encoder + Transformer decoder)** [Ongoing]
 - Built a pipeline using a CNN encoder and Transformer decoder to generate Chest X-ray reports.
 - Normalized medical text, built vocabulary and prepared dataset splits.
 - Trained and evaluated model performance using BLEU, ROUGE and METEOR metrics.

Technical Skills

- **Programming Languages:** C, C++, Java, Python
- **Web Development:** HTML, PHP, JavaScript, Java, Django, Bootstrap
- **UI/UX Design:** Figma
- **Database Management:** MySQL, Microsoft SQL Server, PostgreSQL, MariaDB
- **Data Science & AI/ML:** Python, NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, Keras
- **Data Integration & BI Tools:** Pentaho

Awards & Scholarships

- Received an 80% tuition waiver by the university for consistent academic performance.

Leadership & Volunteering

- Head of Media Team, Southeast Computer Club - led media operations and organized multiple events.
- Volunteered at university events including convocations, orientations and other events, assisting with logistics, coordination and participant management.

Reference

Mushfiquir Rahman Chowdhury

Senior Engineer

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Pankaj Barua

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